

Standard Operating Procedure (SOP) Whole Genome NGS Pipeline (wf_ngs_pipeline.wdl)

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February 9, 2026

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1 Purpose

This SOP describes how to run and interpret results from the `wf_ngs_pipeline.wdl` workflow from the GitHub repository:

- https://github.com/ranlib/whole_genome

The workflow processes paired-end Illumina whole-genome sequencing (WGS) data and performs:

- Quality control and read trimming
- Taxonomic classification
- Reference alignment
- Coverage and QC metrics
- Variant calling and annotation
- Summary reporting with MultiQC

2 Pipeline Overview

The pipeline is written in Workflow Description Language (WDL) and processes multiple samples in parallel. Each sample passes through the following stages:

1. Raw read QC
2. Read trimming and filtering
3. Post-trimming QC
4. Taxonomic classification
5. Alignment to reference genome
6. BAM metrics and whole-genome QC
7. Coverage estimation
8. Variant calling and structural variant detection
9. Variant annotation
10. Aggregated reporting

3 Workflow Steps

3.1 Pre-Alignment Quality Control

- **SeqKit**: Summary statistics of raw reads.
- **FastQC**: Quality metrics on raw FASTQ files.
- **Fastp (Loose & Tight modes)**: Adapter trimming and quality filtering.
- **SeqKit (post-cleaning)**: Statistics after trimming.
- **FastQC (post-cleaning)**: QC reports on cleaned reads.

3.2 Taxonomic Classification

- **Centrifuge**: Classification of reads to identify contamination or microbial composition.

3.3 Alignment and BAM Processing

- **Minimap2**: Alignment of reads to reference genomes.
- **SAMtools**: Indexing reference FASTA and BAM files.
- **Picard / SAMtools metrics**: Alignment and insert-size metrics.

3.4 Coverage and Whole-Genome QC

- **GATK WGS Metrics**: Coverage and quality metrics.
- **Mosdepth**: Depth of coverage statistics.

3.5 Variant Calling and Annotation

- **GATK**: Variant calling producing VCF files.
- **Delly**: Structural variant detection.
- **VCF concatenation**: Merge SNVs and SVs.
- **SnpEff**: Variant effect annotation.

3.6 Summary Reporting

- **MultiQC**: Aggregates all QC, alignment, coverage, and variant reports into a single HTML report.

4 Inputs

Parameter	Description
reads1	Forward FASTQ files
reads2	Reverse FASTQ files
samplenames	Unique sample identifiers
references	Reference genome FASTA files
dockerImages	Map of Docker images for tools
threads	Number of CPU threads
memory	Memory allocation per task
disk_size	Disk allocation for large tasks

5 Outputs

Output	File Type
Cleaned reads	FASTQ
FastQC reports	HTML, ZIP
Centrifuge classification	TSV
Alignments	BAM, BAI
Coverage reports	TXT, TSV

Variant calls	VCF, VCF.GZ
Annotated variants	VCF
MultiQC report	HTML

6 Execution Instructions

Example command using Cromwell:

```
java -jar cromwell.jar run wf_ngs_pipeline.wdl -i inputs.json
```

7 Best Practices

- Validate FASTQ file integrity before running the pipeline.
- Ensure reference genomes are properly indexed.
- Verify Docker images are accessible.
- Adjust memory and CPU resources based on dataset size.
- Review MultiQC output for global QC assessment.

8 Troubleshooting

- **Workflow fails at startup:** Check all required inputs are provided in `inputs.json`.
- **Out-of-memory errors:** Increase memory settings for alignment or variant calling tasks.
- **Empty MultiQC report:** Ensure all upstream tasks completed successfully.

9 References

- https://github.com/ranlib/whole_genome
- <https://openwdl.org>
- <https://github.com/broadinstitute/cromwell>